The ICPP Protocol
A Platform for P2P Privacy-Enabled ICP Transfers

Brief Introduction and Usage Guide

v2.1.00

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Welcome to ICPP

This document explains the operation of ICPP, a new privacy protocol on the Internet Computer, in a non-technical and intuitive way. ICPP denotes privacy-enabled ICP.

Users of the Internet Computer (ICP) infrastructure and ICP token holders have in ICPP a means to execute P2P transactions using high-end cryptography (e.g., encrypted transport tunnels and capsules) alongside other unique features. ICPP also incorporates ephemerality. **Competing protocols are only just beginning to explore this approach for privacy, but ICPP makes the technology available today to the ICP community.**

What does ephemerality mean? In our context, it refers to using short-lived infrastructure layers or code components (in the case of the ICP, canisters) to process P2P transactions.

To understand this, let's use an analogy. Alice needs to cross a river to meet Bob, who is waiting on the opposite shore. She finds a bridge and crosses it, but as soon as she reaches the other side, the bridge vanishes. She cannot retrace her steps and the physical evidence of her crossing has disappeared. No bridge remains, nor is there any visible trace suggesting how Alice could have plausibly reached the other side.

Is ICPP a new Token?

That’s the next, and quite relevant, question many potential users would be inclined to ask, for which there is a simple and direct answer.

Transactions using ICPP involve vanilla ICP tokens.

It means there is no need for an ad-hoc token, or for wrapped or bridged variants. Just plain ICP. The platform allows anyone with an ICCP account to make shielded P2P ICP transfers to other ICCP account holders. **The beneficiary will see ICP tokens, identical to every other ICP token, credited in their account and ready for use.**

The upside?

Transactions on ICPP leave no meaningful traces. At most, only encrypted breadcrumbs.

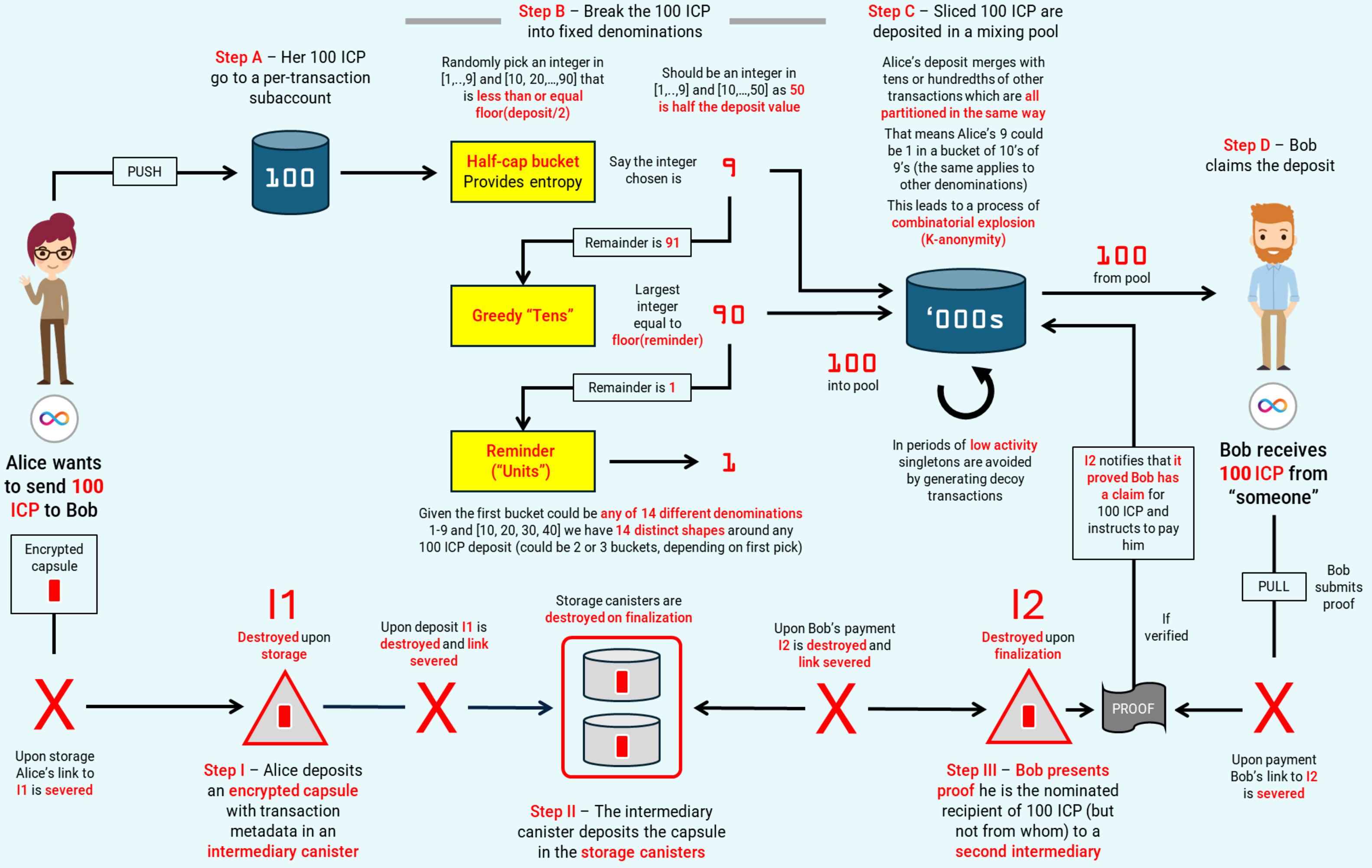
How, you may ask?

The mechanics, in plain language

Let’s say Alice wants to transfer some ICP to Bob.

- The canisters processing P2P transactions, which are spawned on a per-transaction basis, are compartmentalized. **No single canister ever holds both Alice and Bob's identities simultaneously.**
- Besides a Router, a canister Factory, and a Noticeboard canister, each holding no information fully connecting Alice with Bob, **all other canisters intervening in an ICPP-mediated transaction are ephemeral.** Once the transaction completes, they are destroyed. **The partial knowledge each held is gone.** Even if it were possible to reconstruct any canister's state, none ever contained the full picture simultaneously.**The link between Alice and Bob never persisted in any single place.**
- **Transactions are pooled.** Alice’s transfer to Bob is split into pieces and mixed with everyone else’s transfers. Bob withdraws a single amount. An observer sees fragments going in and whole amounts coming out. **Matching them requires guessing by brute force which fragments belong together.**
- **Random time decorrelation** also contributes to hardening privacy, simply because Bob has to action (within a time window) upon knowledge of the transfer.

But how does it work, actually?



ICPP utilizes an infrastructure of permanent and ephemeral canisters to **decouple deposits from claims.** Through advanced cryptography and algorithmics, transaction data is processed so **the Protocol never associates a sender with their recipient.** In this way, ICPP preserves absolute privacy and on-chain unlinkability.

ICPP uses ICP's public ledger but no information is leaked linking Alice to Bob

[1] See https://arxiv.org/abs/2512.23535

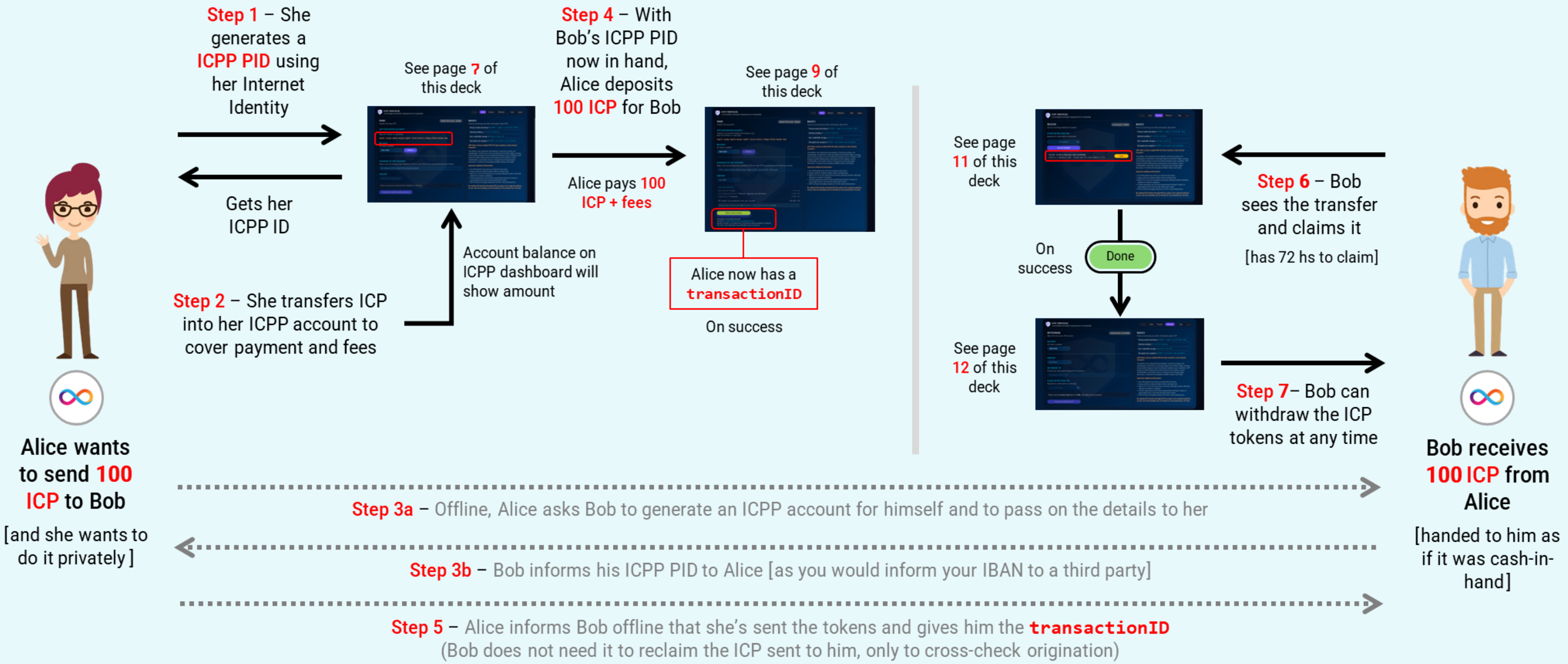
Why ephemerality?

Ephemerality (infrastructure components that are destroyed) **bounds the adversary's computational window.** But it's also a **cryptographic property**, not just operational hygiene.

- Traditional crypto assumes an adversary captures ciphertext, has unlimited time to attack, and security must hold indefinitely or until key expiry. **Because encrypted transactions are stored forever on-chain.**
- In comparison, an adversary must attack IPCC during a **very limited canister lifetime**, leaving no persistent cyphertext to attack offline, and no keys to exfiltrate later.
- **This is forward secrecy applied at the architectural level**. Even if adversary compromises the system **after transaction completion**, they can't decrypt past transactions because the ephemeral state is cryptographically gone.
- As no single component in IPCC holds complete information, **an attacker must compromise multiple canisters simultaneously**. Doing so **under time pressure** (which compounds across components).
- To recap: IPCC is a **time-bounded, compartmentalized key agreement protocol** that leverages the IC unique properties and cryptography to create privacy guarantees that traditional blockchain privacy protocols cannot achieve.

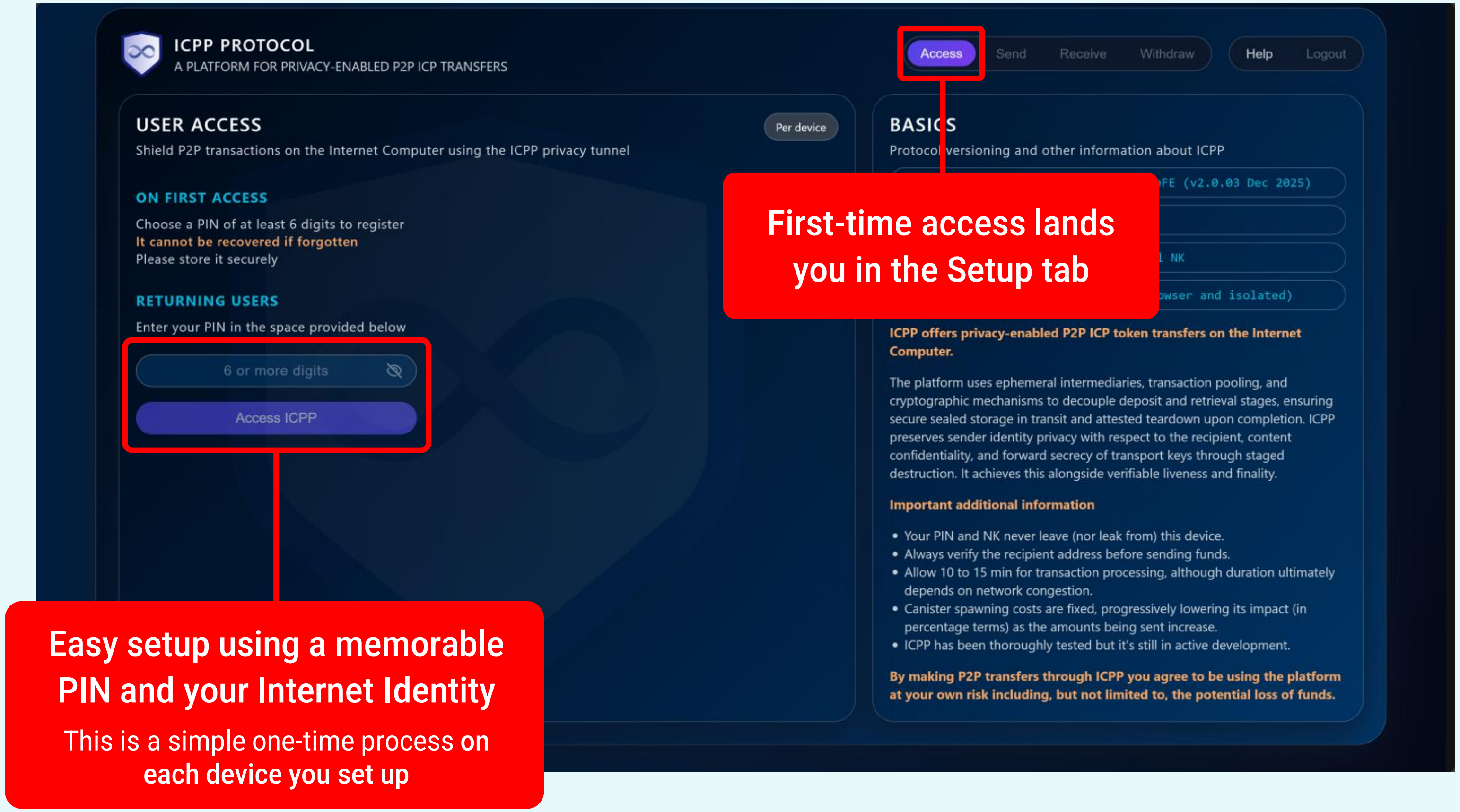
Anatomy of a Transaction using ICPP

No need for command-based instructions. A visual interface makes it **simple and stress-free.** Online at https://icpp.tech



The ICPP Platform UX

We designed the ICPP interface to be as **easy, intuitive, and secure** as possible.



For access, go to https://icpp.tech hosted on the Internet Computer.

After choosing a PIN you will be directed to DFINITY Internet Identity gateway.

Proceed to link your IC account to ICPP. If successful, you will be redirected to the dashboard.

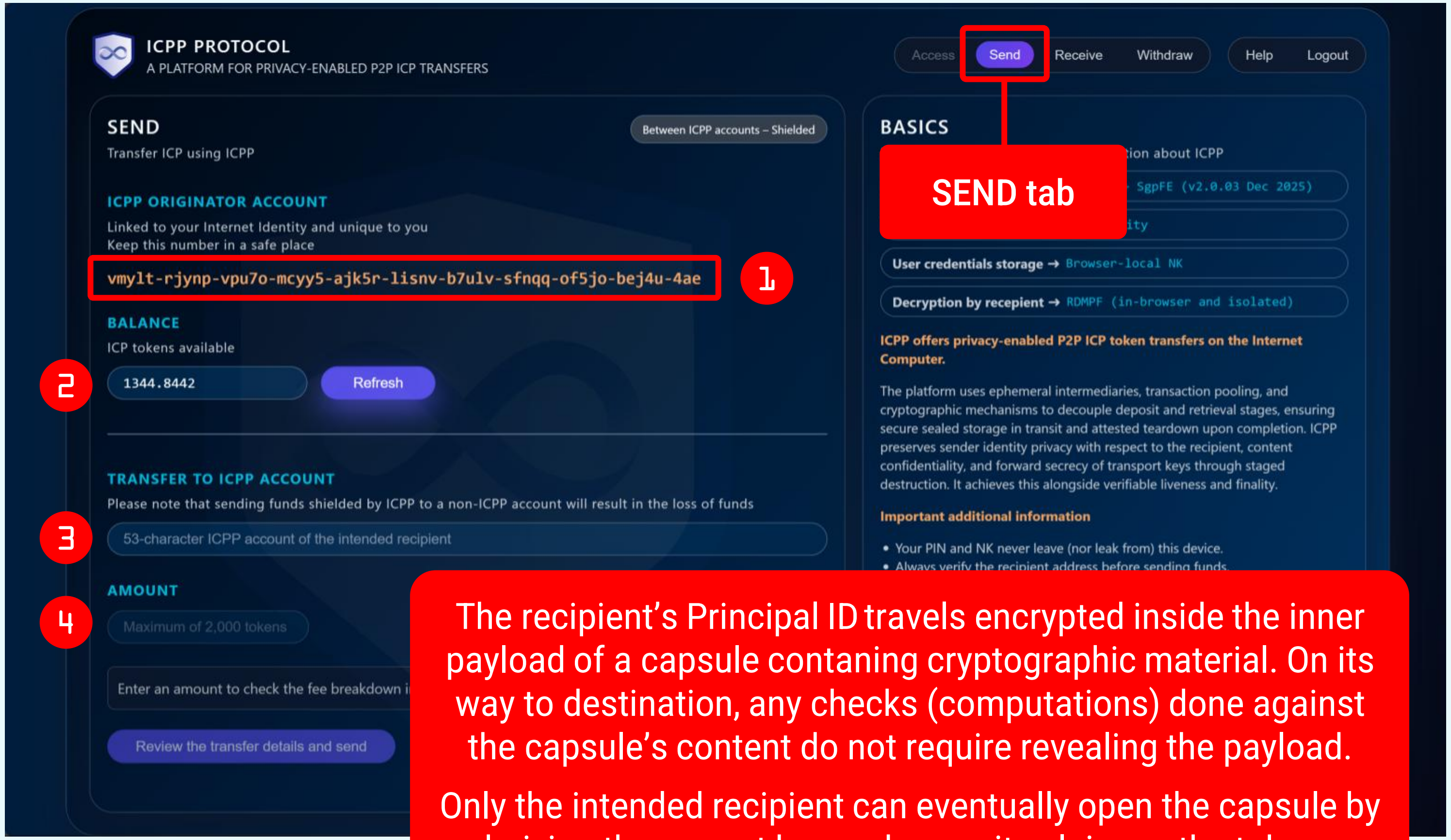
Credentials are device and browser specific. The combo generates a locally-stored encrypted NK that gates access to the platform.

Important: The ICPP dashboard only works in a desktop or laptop devices.

The ICPP Platform UX

Sending ICP using ICPP is **quite simple**
It involves providing some **basic information** common to any assets transfer

1. This is your **unique ICPP account ID**
   [Transfer to this ICRC-1 account enough ICP tokens to provision the transactions you intend to make]
2. **Balance** of your ICPP account
   [ICP tokens available to use]
3. **Destination** for the ICP transfer
   [ICPP account of the intended recipient]
4. The **amount of ICP tokens** to send
   [up to 2,000 ICP per transaction]

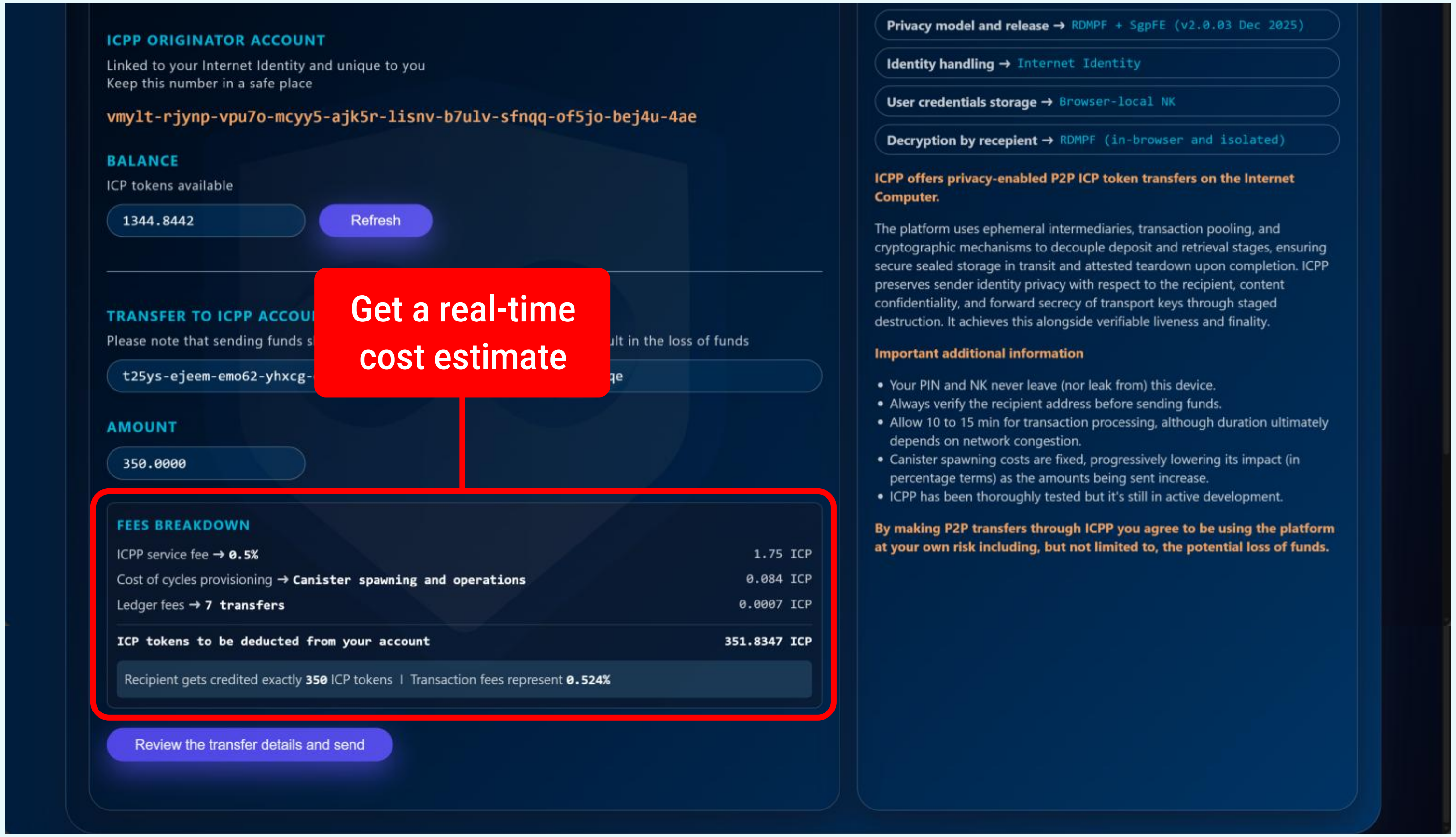


The recipient's Principal ID travels encrypted inside the inner payload of a capsule containing cryptographic material. On its way to destination, any checks (computations) done against the capsule's content do not require revealing the payload.

Only the intended recipient can eventually open the capsule by deriving the correct key and prove its claim on the tokens.

The ICPP Platform UX

Complete transparency on transaction costs



ICPP adopts a Wise-style pricing structure. **The sender pays platform fees and operational costs** (canister spawning and ledger fees).

The **beneficiary** receives the **net amount of ICP tokens** being sent.

Plaform fees are

0.5%

For the time being, ICPP sets a **cap** of **2,000 ICP tokens per transaction** to prevent abuse, whale crowding-out, and money laundering.

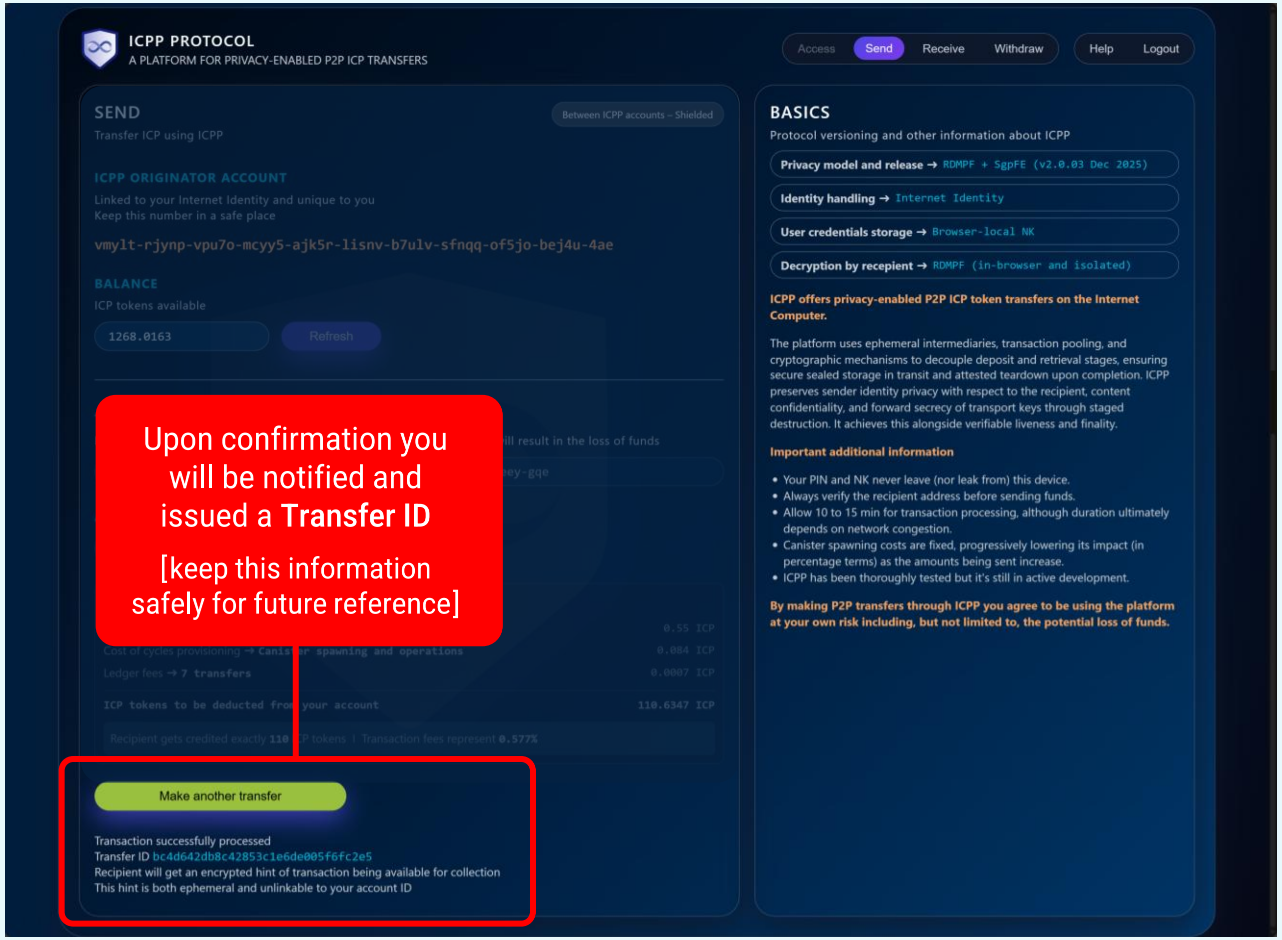
The ICPP Platform UX

Outbound transactions demand about **10 to 15 minutes** for confirmation (sometimes less)

Transactions on ICPP are final

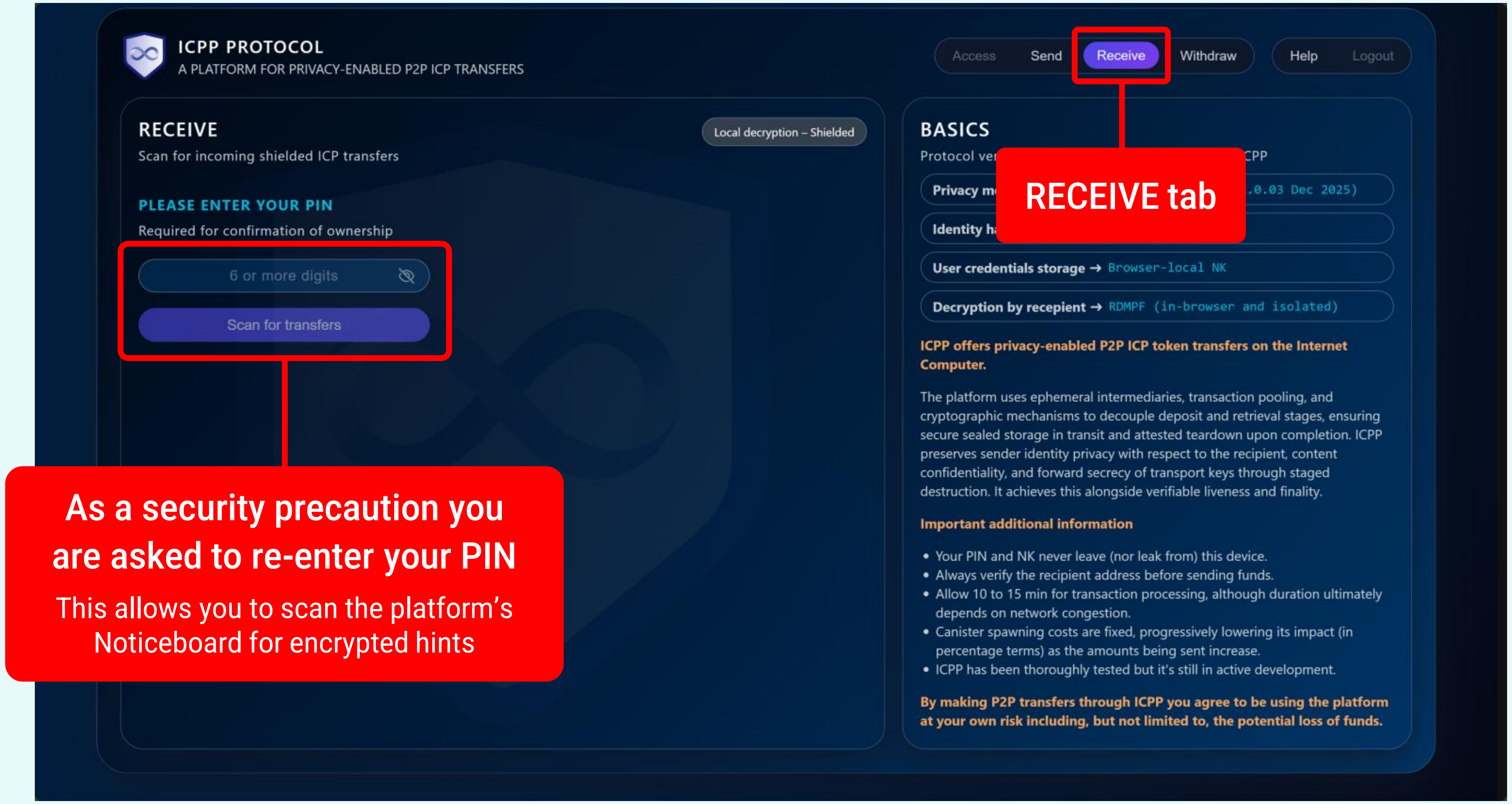
Once a transaction is executed, the only way to claw back the tokens is to ask the recipient for their reimbursement.

[we are currently exploring mechanisms for reverting transactions without compromising privacy]



The ICPP Platform UX

Receiving ICP tokens using ICPP is equally **straightforward**



Simply **provide your PIN** for the dashboard to **scan the ICPP Noticeboard** containing hints.

Users **attempt decryptions** on those announcements (using an efficient algorithm). Success identifies which transaction(s) have you as beneficiary.

The reclaim process is very easy.

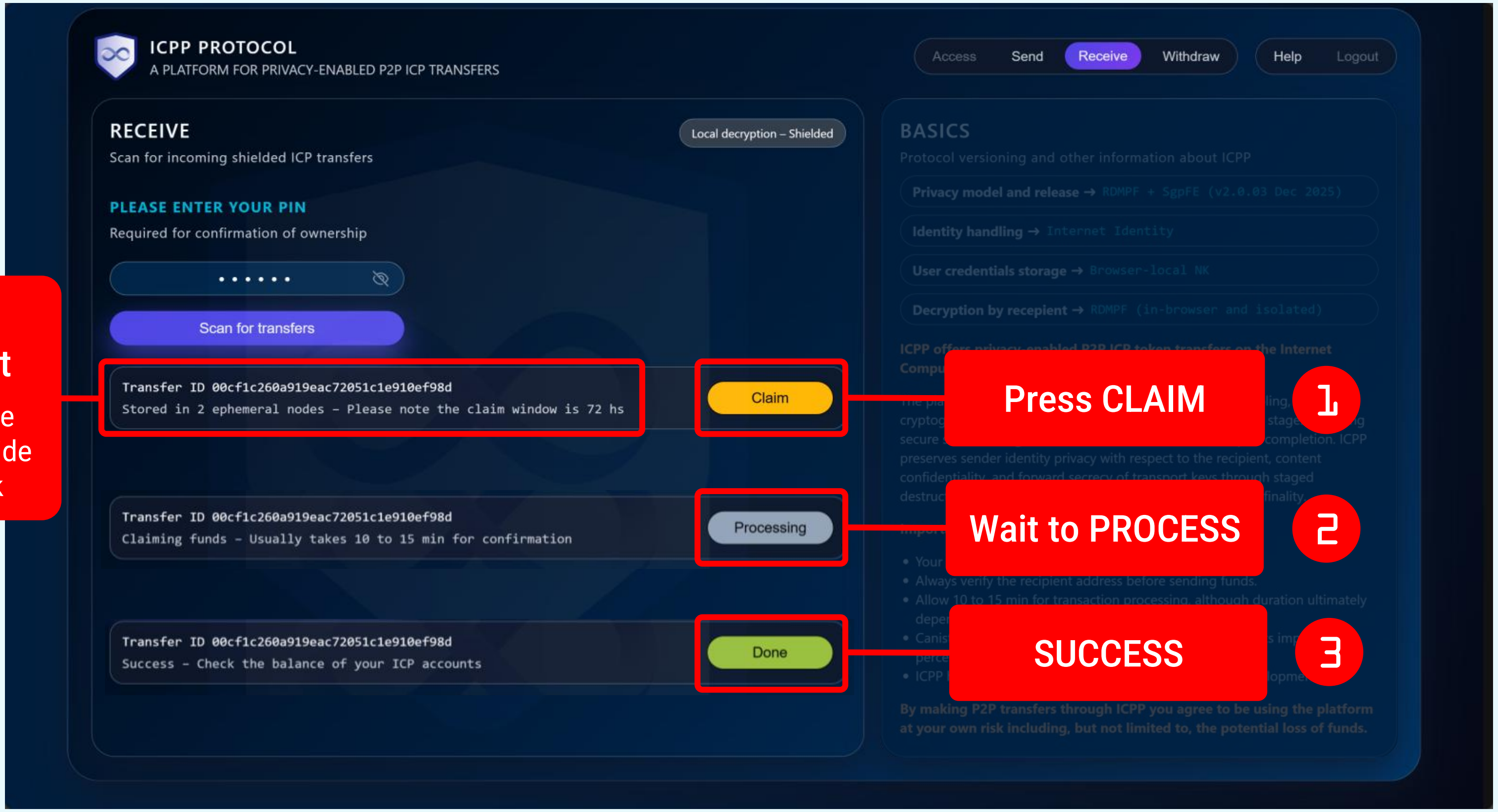
Similarly to when you send ICP using ICPP, when claiming funds the process can take **10 to 15 minutes** (depending on network congestion).

The ICPP Platform UX

Now just wait for any ICP token sent to you to land on your IC account

If you have a transfer pending you can CLAIM it

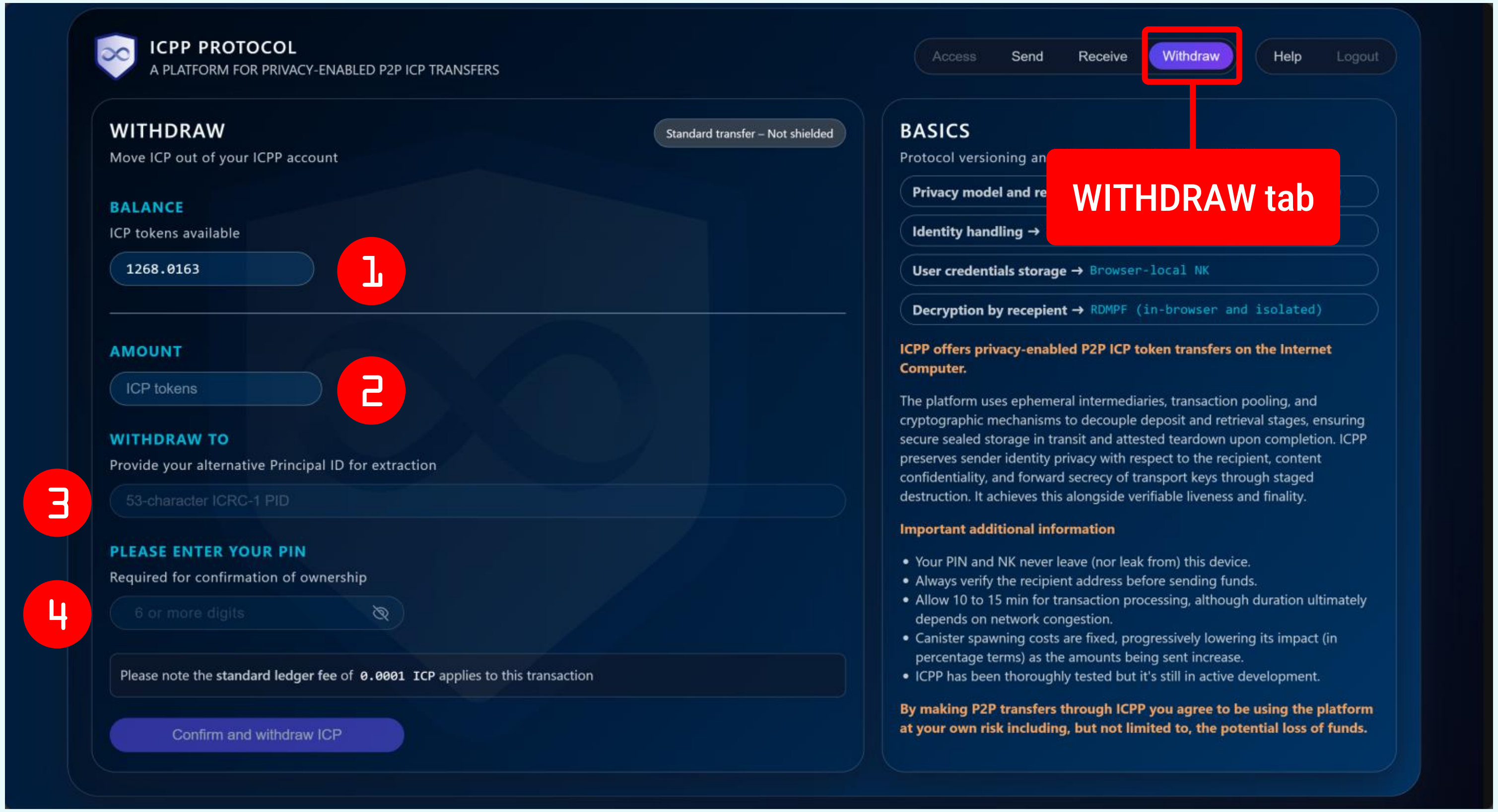
Ask the sender to provide you the Transaction ID [using an offline side channel] for you to cross-check



Important: On very rare instances a transaction could get stuck. These include (but are not limited to) loss of connection, II identity and delegation staleness, or IC network hiccups. The platform has several in-built failure prevention features. As first port of call, wait until the TRY AGAIN option becomes available. If that doesn't work, restart your browser, login and rescan. Operationally, if you left the IPCC dashboard sitting idle for more than 30 min we recommend to log out and then back in to ensure delegation is refreshed before attempting any action.

The ICPP Platform UX

Withdraw ICP tokens on ICPP to **your account of choice** for immediate use



1. **Balance** of your ICPP account
   [ICP tokens available for extraction]
2. Decide on the **amount of ICP tokens** to move out
   [Transfer to any account supporting the ICRC-1 standard]
3. **Recipient** of the ICP tokens from ICPP
   [Account where the tokens will be sent to]
4. **Enter your PIN** and withdraw the tokens
   [with no caps or other restrictions]

Making ICP transactions private on the Internet Computer

https://icpp.tech

icpp.tech@proton.me